

Program:	Diploma in Integrated Circuit Design and Fabrication		
Course Code:	5372B	Course Title:	Semiconductor Packing & Testing
Semester:	5	Credits:	3
Course Category:	Program Elective Course		
Periods per week:	3 (L:3, T:0, P:0)	Periods per semester:	45

Course Objectives:

- To gain knowledge of the semiconductor packaging hierarchy and its role in protecting and connecting electronic components.
- To analyze material properties (ceramics, polymers, metals) and select appropriate materials for packaging based on thermal, electrical, and mechanical requirement.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic Knowledge in semiconductor physics	2003	Applied Physics II	2
Semiconductor theory, diodes, and transistors	2041	Basic Electronics	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Explain the basics of electronic systems packaging.	11	Understanding
CO2	Summarize electrical issues and design process in systems packaging.	10	Understanding
CO3	Explain various chip-level packaging technologies and processes.	11	Understanding

CO4	Explain various aspects of design for reliability and electrical testing of semiconductor packaging.	11	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the basics of electronic systems packaging.		
M1.01	Explain electronic packaging.	2	Understanding
M1.02	Describe the packaging hierarchy	2	Understanding
M1.03	Explain different types of IC packaging.	3	Understanding
M1.04	Identify materials used in microelectronic packaging	4	Understanding
Contents: Overview of Electronic Systems Packaging: Functions of Electronic Packaging, Packaging Hierarchy. IC packaging: MEMS packaging, consumer electronics packaging, medical electronics packaging, Trends and Challenges Materials for Microelectronic packaging, Packaging Material Properties, Ceramics, Polymers, and Metals in Packaging, Material for high density interconnect			

CO2	Summarize electrical issues and design process in systems packaging.		
M2.01	Explain various electrical issues in systems packaging	6	Understanding
M2.02	Describe design process for electrical packaging	4	Understanding
	Series Test – I	1	
Contents Electrical Issues in Packaging: Electrical Issues of Systems Packaging, Signal Distribution, Power Distribution, Electromagnetic Interference, Transmission Lines, Clock Distribution, Noise Sources. Design Process: Electrical Package Design. Interconnect Capacitance, Resistance and Inductance fundamentals.			
CO3	Explain various chip-level packaging technologies and processes.		
M3.01	Explain the purpose and requirements of IC assembly	1	Understanding
M3.02	Explain various technologies used in IC assembly	5	Understanding
M3.03	Summarize single-chip packaging	3	Understanding
M3.04	Summarize multi-chip packaging (MCP)	2	Understanding
Contents Chip Level Packaging: IC Assembly - Purpose, Requirements, Technologies, Wire bonding, Tape Automated Bonding, Flip Chip, Wafer Level Packaging, reliability, wafer level burn – in and test. Single chip packaging: functions, types, materials processes, properties, characteristics, trends. Multi-chip packaging: types, design, comparison, trends.			
CO4	Explain various aspects of design for reliability and electrical testing of semiconductor packaging.		
M4.01	Explain the fundamentals of design for reliability in electronic packaging.	1	Understanding
M4.02	Explain the various failure mechanisms in electronic packaging.	4	Understanding

M4.03	Outline accelerating factors in failure mechanisms.	2	Understanding
M4.04	Explain the process and methods involve in electrical testing	4	Understanding
	Series Test – II	1	
Contents Design for Reliability: Fundamentals, Failure mechanisms, Thermal mismatch and fatigue – failures – thermo mechanically induced –electrically induced – chemically induced. Accelerating factors and design solutions (Basics only). Electrical Testing: System level electrical testing, Interconnection tests, Active Circuit Testing, Design for Testability			

Text/Reference

T/R	Book Title/Author
T1	Tummala, Rao R., Fundamentals of Microsystems Packaging, McGraw Hill, 2001.
T2	Tummala, Rao R, Microelectronics packaging handbook, McGraw Hill, 2008
T3	Blackwell (Ed), The electronic packaging handbook, CRC Press, 2000
R1	Bosshart, Printed Circuit Boards Design and Technology, TataMcGraw Hill, 1988.
R2	R.G. Kaduskar and V.B.Baru, Electronic Product design, Wiley India, 2011

Online resources:

Sl.No.	Website Link
1	https://www.essc-india.org/semicon/assembly-&-packaging.php
2	https://semiengineering.com/knowledge_centers/packaging/